

RDF Funded Research Project: Budget and Expected Outcomes

This form is to be completed by the Principal Investigator (PI) after the award has been approved by the XJTLU Research Review Panel. Please present your APPROVED BUDGET and your EXPECTED OUTCOMES in this form.

RDF Ref No.: RDF-24-02-031

Principal Investigator			
Name	Izhar Oswaldo Escudero Ornelas	School	School of Internet of Things
		Department	
Project Information			
Project Title	Real-Time Human Coordination and Predictive Maintenance in Multi-Agent Systems		
Start Date	01/07/2025	Complete Date	30/06/2027
Awarded Amount (RMB)	100,000		

Approved Budget		
Please paste the approved budget of the project here.		
Categories	Amount (RMB)	Justification
Equipment Expenses	75,000	High-end computer: CPU (Intel i9), GPU (NVIDIA RTX 4080), 64GB RAM, 1TB SSD (24,000 RMB). Critical for running ABMS/RL simulations with low latency. Existing lab machines lack computational power for large-scale simulations. While the University's HPC excels at batch processing for large-scale model training and simulations (back-end computation), the requested equipment is essential for addressing latency-sensitive, domain-specific requirements inherent to industrial cyber-physical systems, particularly for real-time data processing near the milling machine. Benchtop CNC Milling Machine: 3-axis Tormach PCNC 440 (36,000 RMB). Critical for real-world predictive maintenance testing and data collection. Sensors (vibration, thermal), tooling kits, and spare parts for milling machine maintenance experiments (15,000 RMB). Sensors enable data collection on machine degradation, which is foundational for training the predictive maintenance AI model.
Operational Expenses	5,800	Industry visits: 2 trips to Suzhou Industrial Park partners (transportation: 2,000 RMB). Pilot testing with partners like Huawei or Haier requires on-site integration of the AI framework with their production

		lines. These visits are critical for refining the model's adaptability to real-world manufacturing environments. Contingency for unforeseen expenses (e.g., component replacements, additional cloud costs) (3,800 RMB).
Labour Expenses	19,200	Research Assistant (RA): The project requires dedicated support for critical tasks that are time-intensive and demand meticulous attention to detail. Hiring an RA ensures efficient execution of foundational work, allowing the core research team to focus on higher-level strategic and technical challenges. The RA's contributions will directly enhance the quality and speed of the project's deliverables. Responsibilities: 1. Data Preprocessing: Clean, organize, and prepare datasets for model training and analysis, ensuring data quality and consistency. 2. Simulation Setup: Configure and run simulations based on project requirements, including parameter tuning and result documentation. 3. Literature Review: Conduct comprehensive reviews of academic and industry publications to identify relevant methodologies, tools, and best practices. Impact: The RA's work will streamline the project's foundational processes, enabling faster progress and higher-quality outcomes. Their contributions in data preprocessing, simulation setup, and literature review will provide essential support for the project's success. Workload: The RA will work 4 months $\times$ 40 hours/month $\times$ 60 RMB/h consultants to support their work, totaling 9,600 RMB. This workload ensures sufficient time to complete tasks thoroughly while maintaining flexibility to address emerging project needs. Consultants: The project involves hiring specialized consultants for the integration and optimization of the AI framework. These consultants bring expertise in industrial cyber-physical systems and real-time data processing, ensuring seamless implementation and alignment with partner requirements. The estimated cost for consultant fees is (2,400 RMB per day, requiring at least 4 days = 9,600 RMB), covering their contributions to model refinement, system integration, and technical support during pilot testing phases.
Total amount:		100,000RMB

Timeline and Work Plan

No.	Timeline	Work Plan
1	Jul 2025 – Dec 2025	Finalize hybrid MAS architecture; identify tools, datasets, and simulation parameters.
2	Jan 2026 – Jun 2026	Build an ABMS prototype (AnyLogic/NetLogo) and define agent roles (AI, humans, machines)
3	Jul 2026 – Dec 2026	Train RL algorithms (TensorFlow) for adaptive decision-making; establish real time data workflows.
4	Jan 2027 – Mar 2027	Develop human-AI dashboard; conduct usability tests with operators
5	Apr 2027 – May 2027	Deploy system in milling machine simulation; validate accuracy, downtime reduction
6	Jun 2027	Analyze results; publish findings; present to academic/industry stakeholders.

Outcomes Summary				
	Total Number		Number	Note
Publication	4	Journal Paper	2	2 journal papers submitted to high-impact journals (IEEE Transactions on Industrial Informatics, Engineering Applications of Artificial Intelligence) by Year 2, focusing on hybrid ABMS-RL framework validation and human-in-the-loop optimization
		Conference Paper	1	1 conference paper presented at top-tier venues (e.g., IEEE CASE, NeurIPS) by Year 1, showcasing real-time adaptability and industrial case

				studies.
		Other (e.g. book, chapter, monograph)	1	1 technical book chapter (e.g., AI-Driven Industrial Automation) submitted to a Springer or IEEE Press series by Year 3, synthesizing methodological innovations and sectoral applications
Intellectual Property		Invention	0	
		Utility Model	0	
		Design Patent	0	
		Software Copyright	0	
Practice Outcome		Visual Art Work	0	
		Design, Architecture, Interiors	0	
		Textual Work	0	
		Audio-Visual Recording	0	
		Inter-Arts	0	
		Digital Creative Work	0	
		Website / Web-Based Exhibition	0	
Exhibition or Presentation		Competition entries	0	
		Physical or Live Exhibition	0	
		Exhibition via Virtual/Online/Broadcast/Projection	0	
		Inclusion within School or University curriculum	0	
		Research Report	0	
Others (e.g. social Impact, host events)	(Please give description here.) Hosting knowledge-sharing events at the university, such as workshops, seminars, and public lectures, to disseminate research findings, foster collaboration, and engage the academic and industrial communities in discussions on the project's innovations and applications			
What external funding will you attempt to seek?	Funding body (e.g. *NSFC)		National Grants: o National Natural Science Foundation of China (NSFC): Key programs like General Program or Key Projects for AI-driven	

		industrial innovation. o Ministry of Science and Technology (MOST): National Key R&D Program (e.g., Intelligent Robots, Advanced Manufacturing). o Provincial/Municipal Funds: Jiangsu Province or Suzhou Municipal grants (e.g., Suzhou Science and Technology Plan) supporting Industry 4.0 and AI integration.
	Programme (e.g. General Programme)	
	Expected submission date	

*NSFC refers to National Nature Science Foundation of China

Ethical Requirements of Research Projects Involving Human and Animals

Research which involves human (including human participants, survey respondent, human genetic resources and personal data) or animals will always require formal ethical consideration.

- ☒ Yes
☐ No

Declaration:

I accept the XJTLU RDF fund for my proposed research project.

By making use of the University Research Development Fund, I will do my best

- to ensure that the research resources are used for the ultimate benefits of the University in enhancing his/her research profiles and in generating maximum research outcome contributing to knowledge generation, innovation and benefiting the University and the wider Society;
- to conduct the research in line with Good Research Conduct and in Observance of Professional Guidance and standards, research ethics as well as the University related policies and procedures;
- to complete the RDF funded research project as outlined in the Research Proposal and deliver the expected outcomes as specified above.

Signature of PI: **Izhar Oswaldo Escudero Ornelas**

Date: **04 June,2025**

Stamp of Research Management Office

Date: **01 April,2025**

